

LoRC: Low-rank Quantization Correction

1 Introduction

We consider the problem of compressing a pretrained language model f_θ while preserving its performance on a target domain $\mathcal{D}_{\text{target}}$, potentially at the expense of out-of-domain generality. Our approach—**Low-rank Quantization Correction**—identifies low-dimensional subspaces in which activation statistics differ between domains and uses these subspaces to construct low-rank corrections to quantized weights.

The method is motivated by the observation that quantization error is not equally important in all activation directions. If a domain frequently occupies a particular subspace, errors in the corresponding weight directions can have a disproportionately large effect on the domain’s output.

1.1 Covariance estimation

For a linear layer with weight

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}},$$

we use column-vector activations $x \in \mathbb{R}^{d_{\text{in}}}$ and $y \in \mathbb{R}^{d_{\text{out}}}$, with

$$y = Wx.$$

For each domain $\mathcal{D}$, we estimate the covariance matrices

$$C_{\text{pre}}^{\mathcal{D}} = \mathbb{E}[xx^\top] - \mathbb{E}[x]\mathbb{E}[x]^\top \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}, \quad (1)$$

$$C_{\text{post}}^{\mathcal{D}} = \mathbb{E}[yy^\top] - \mathbb{E}[y]\mathbb{E}[y]^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{out}}}. \quad (2)$$

Given samples $\{x_i\}_{i=1}^N$ from a domain, the corresponding population-normalized estimator is

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i, \quad (3)$$

$$\hat{C} = \frac{1}{N} \sum_{i=1}^N x_i x_i^\top - \hat{\mu} \hat{\mu}^\top. \quad (4)$$

The normalization is performed independently for each domain so that differences in the number of collected samples do not affect the estimated covariance.

1.2 Domain subspace extraction

Given two domains $\mathcal{D}_1$ and $\mathcal{D}_2$, we form the difference between their covariance matrices:

$$\Delta C = C^{\mathcal{D}_1} - C^{\mathcal{D}_2}. \quad (5)$$

In exact arithmetic, both covariance matrices are symmetric, and therefore ΔC is symmetric. For numerical estimates, we explicitly symmetrize:

$$\Delta C \leftarrow \frac{1}{2} (\Delta C + \Delta C^\top). \quad (6)$$

More generally, weighted covariances can be used:

$$\Delta C = \alpha C^{\mathcal{D}_1} - \beta C^{\mathcal{D}_2}, \quad \alpha, \beta \geq 0. \quad (7)$$

In this case the eigenvalues measure differences in *weighted* activation variance. Unless there is a specific normalization objective, we use $\alpha = \beta = 1$.

We compute the eigendecomposition

$$\Delta C = Q \Lambda Q^\top, \quad (8)$$

where Q is orthogonal and Λ is diagonal.

For any unit vector v ,

$$v^\top \Delta C v = v^\top C^{\mathcal{D}_1} v - v^\top C^{\mathcal{D}_2} v. \quad (9)$$

Thus eigenvectors associated with the largest positive eigenvalues identify directions with substantially greater activation variance in $\mathcal{D}_1$, while eigenvectors associated with the most negative eigenvalues identify directions with substantially greater activation variance in $\mathcal{D}_2$.

We retain the K largest positive eigen-directions and the K most-negative eigen-directions, subject to their availability, forming column-orthonormal matrices

$$V_1, V_2 \in \mathbb{R}^{d \times K}.$$

The resulting subspaces describe differences in activation covariance; they do not by themselves imply that quantization error is concentrated in those directions.

1.3 Connection to quantization error

Let W_{base} denote the dequantized NF4 weight and define the quantization error

$$E = W - W_{\text{base}}, \quad (10)$$

$$W_{\text{base}} = Q^{-1}(Q(W)). \quad (11)$$

For a centered input activation x with covariance C , the output error caused by E is

$$\delta y = Ex.$$

Its expected squared norm is

$$\mathbb{E} [\|\delta y\|_2^2] = \mathbb{E} [x^\top E^\top Ex] \quad (12)$$

$$= \text{tr} (ECE^\top). \quad (13)$$

Consequently, the difference in expected quantization-error energy between two domains is

$$\Delta L_E = \text{tr} (E(C^{\mathcal{D}_1} - C^{\mathcal{D}_2})E^\top) \quad (14)$$

$$= \text{tr} (E\Delta CE^\top). \quad (15)$$

This provides a direct connection between domain-dependent activation statistics and the effect of quantization. In particular, activation subspaces are useful for correction when the corresponding directions also contain significant quantization error.

1.4 Correction factors

Consider first an input-space domain subspace

$$V \in \mathbb{R}^{d_{\text{in}} \times K}, \quad V^\top V = I.$$

The component of the quantization error acting on this subspace is

$$E_{\text{proj}} = E(VV^\top). \quad (16)$$

Rather than constructing the full $d_{\text{in}} \times d_{\text{in}}$ projection matrix, define

$$A = EV \in \mathbb{R}^{d_{\text{out}} \times K}. \quad (17)$$

Then

$$E_{\text{proj}} = AV^\top. \quad (18)$$

Compute the compact SVD

$$A = U_A \Sigma_A R_A^\top. \quad (19)$$

Keeping the first k singular components, where

$$k = \min(K, d_{\text{out}}, d_{\text{in}}),$$

gives

$$A \approx U_{A,k} \Sigma_{A,k} R_{A,k}^\top. \quad (20)$$

Therefore

$$E_{\text{proj}} \approx U_{A,k} \Sigma_{A,k} R_{A,k}^\top V^\top \quad (21)$$

$$= U_{A,k} \Sigma_{A,k} (V R_{A,k})^\top. \quad (22)$$

Define

$$V_{\text{act}} = (V R_{A,k}) \Sigma_{A,k}^{1/2} \in \mathbb{R}^{d_{\text{in}} \times k}, \quad (23)$$

$$U_{\text{write}} = U_{A,k} \Sigma_{A,k}^{1/2} \in \mathbb{R}^{d_{\text{out}} \times k}. \quad (24)$$

The resulting low-rank approximation satisfies

$$E_{\text{proj}} \approx U_{\text{write}} V_{\text{act}}^\top. \quad (25)$$

The factors are stored in bfloat16.

For an output-space domain subspace

$$V \in \mathbb{R}^{d_{\text{out}} \times K},$$

the corresponding projection is

$$E_{\text{proj}} = (V V^\top) E. \quad (26)$$

The analogous low-rank factorization can be obtained by applying the same procedure to $E^\top$ and transposing the resulting factors.

1.5 Hybrid forward pass

For a row-vector representation of the input activation,

$$x \in \mathbb{R}^{1 \times d_{\text{in}}},$$

the base forward pass is

$$z_{\text{base}} = x W_{\text{base}}^\top. \quad (27)$$

Using the low-rank approximation

$$E_{\text{proj}} \approx U_{\text{write}} V_{\text{act}}^\top,$$

the corresponding output correction is

$$z_{\text{corr}} = x E_{\text{proj}}^\top \quad (28)$$

$$\approx x V_{\text{act}} U_{\text{write}}^\top \quad (29)$$

$$= (x V_{\text{act}}) U_{\text{write}}^\top. \quad (30)$$

Thus the hybrid forward pass is

$$\boxed{z = x W_{\text{base}}^\top + (x V_{\text{act}}) U_{\text{write}}^\top}. \quad (31)$$

The first term uses the quantized NF4 weight, while the second term uses the small bfloat16 low-rank correction.

1.6 Spectral rank selection

The singular values of

$$A = EV$$

measure the strength of the best orthogonal low-rank components of the projected quantization error. They should not be interpreted as contributions of the individual original domain eigenvectors, because the right singular vectors R_A can rotate and mix those directions.

A simple rank-selection rule is therefore to retain singular values above a threshold:

$$\text{keep}_i = \mathbf{1}(\sigma_i > \tau). \quad (32)$$

If a fixed fraction `keep_pct` of the singular components is desired, the threshold can be selected as

$$\tau = \text{quantile}(\{\sigma_i\}, 1 - \text{keep_pct}). \quad (33)$$

Alternatively, for a fixed rank k , the k largest singular values are retained directly.

This procedure provides a deterministic spectral rank-selection heuristic and does not require gradient-based mask training.

1.7 Disjunction score

To evaluate whether a domain subspace is genuinely domain-specific, we can measure the effect of removing the corresponding subspace from the original weight matrix.

For an input-space subspace V ,

$$W' = W - W(VV^\top). \quad (34)$$

For an output-space subspace V ,

$$W' = W - (VV^\top)W. \quad (35)$$

Let $\Delta\text{PPL}_{\text{own}}$ denote the perplexity change on the domain from which the subspace was extracted, and let $\Delta\text{PPL}_{\text{other}}$ denote the corresponding change on another domain.

A clipped ratio can be defined as

$$D = \frac{\max(\Delta\text{PPL}_{\text{own}}, \varepsilon)}{\max(\Delta\text{PPL}_{\text{other}}, \varepsilon)}, \quad \varepsilon > 0. \quad (36)$$

Large D indicates that the subspace has a substantially larger effect on its source domain than on the comparison domain.

This diagnostic measures domain dependence of the underlying weight directions. It should be distinguished from a direct measurement of the effectiveness of the quantization correction itself. For the latter, one should also measure the change obtained by removing

$$U_{\text{write}} V_{\text{act}}^{\top}$$

from the quantized model.

1.8 Streaming covariance products

Explicitly storing a dense covariance matrix requires $\mathcal{O}(d^2)$ memory. For large transformer layers, it is preferable to estimate covariance–vector products directly.

For a domain with mean $\mu = \mathbb{E}[x]$,

$$Cv = \mathbb{E}[x(x^{\top}v)] - \mu(\mu^{\top}v). \quad (37)$$

This permits iterative eigensolvers to operate without explicitly constructing C .

A low-rank subspace estimate can alternatively be maintained using on-line PCA methods. For example, a simple Oja update is

$$V \leftarrow V + \eta x(x^{\top}V), \quad (38)$$

followed by orthonormalization,

$$V \leftarrow \text{QR}(V). \quad (39)$$

This reduces persistent storage from $\mathcal{O}(d^2)$ to $\mathcal{O}(dK)$, although the computational cost of processing the activation stream remains dependent on dK per sample.

For the two-domain difference, separate covariance products can be estimated and combined as

$$\Delta C v = C^{\mathcal{D}_1} v - C^{\mathcal{D}_2} v. \quad (40)$$

1.9 Randomized and Krylov eigendecomposition

Because ΔC is symmetric but generally indefinite, an eigensolver must be capable of recovering both large positive and large negative eigenvalues.

A Lanczos method is a natural choice because it operates using only matrix–vector products with ΔC .

A randomized range finder can also be used. Draw

$$\Omega \sim \mathcal{N}(0, 1)^{d \times (K+p)} \quad (41)$$

and compute

$$Y = \Delta C \Omega. \quad (42)$$

For improved capture of both ends of the spectrum, optional power iterations can be applied:

$$Y = (\Delta C)^{2q+1} \Omega. \quad (43)$$

Since ΔC is symmetric, the magnitude of an eigenvalue λ_i is amplified as

$$|\lambda_i|^{2q+1},$$

allowing both large positive and large negative eigenvalues to be captured.

After orthonormalization,

$$Q = \text{QR}(Y), \quad (44)$$

form the Rayleigh–Ritz matrix

$$T = Q^\top \Delta C Q. \quad (45)$$

Then compute

$$T = \tilde{U} \tilde{\Lambda} \tilde{U}^\top, \quad (46)$$

and approximate the corresponding eigenvectors by

$$V_{\text{ext}} = Q \tilde{U}. \quad (47)$$

For dense covariance operators, each matrix–block multiplication costs approximately

$$\mathcal{O}(d^2(K + p)).$$

Thus the total cost depends on the number of Krylov or power iterations, but avoids the $\mathcal{O}(d^3)$ cost of a full dense eigendecomposition.

1.10 Complexity of the low-rank correction

For an input-space subspace $V \in \mathbb{R}^{d_{\text{in}} \times K}$, forming

$$A = EV$$

costs

$$\mathcal{O}(d_{\text{out}} d_{\text{in}} K)$$

for a dense error matrix.

The subsequent SVD of

$$A \in \mathbb{R}^{d_{\text{out}} \times K}$$

is substantially cheaper than an SVD of the full $d_{\text{out}} \times d_{\text{in}}$ matrix. For $K \ll d_{\text{in}}, d_{\text{out}}$, its cost is approximately

$$\mathcal{O}(d_{\text{out}} K^2)$$

using a standard dense thin SVD.

The projection matrix $VV^\top$ therefore never needs to be constructed.

If the quantized weight is processed blockwise, the full-precision error matrix E also need not be materialized, allowing the correction to be computed directly from quantized-weight blocks.

1.11 Multi-domain extension

For $D > 2$ domains, pairwise covariance differences can be generalized using Fisher’s linear discriminant framework.

Let

$$\mu_d = \mathbb{E}[x \mid \mathcal{D}_d], \quad \mu = \sum_{d=1}^D \pi_d \mu_d,$$

where π_d is the empirical probability of domain d .

The between-domain covariance is

$$C_{\text{between}} = \sum_{d=1}^D \pi_d (\mu_d - \mu)(\mu_d - \mu)^\top. \quad (48)$$

The pooled within-domain covariance is

$$C_{\text{within}} = \sum_{d=1}^D \pi_d C_d. \quad (49)$$

A generalized Fisher subspace can be obtained from

$$C_{\text{between}} v = \lambda C_{\text{within}} v. \quad (50)$$

Equivalently, one can solve

$$\max_W \quad \text{tr}(W^\top C_{\text{between}} W) \quad (51)$$

$$\text{subject to} \quad W^\top C_{\text{within}} W = I. \quad (52)$$

In high-dimensional activation spaces, C_{within} may be singular or poorly conditioned. A regularized version is therefore preferable:

$$C_{\text{within}}^{(\lambda)} = C_{\text{within}} + \lambda I, \quad \lambda > 0. \quad (53)$$

The resulting generalized eigenvectors provide a single subspace that captures domain-discriminative mean variation. If covariance differences rather than mean differences are the primary signal of interest, a separate covariance-discriminant construction is required; standard Fisher LDA alone does not capture all such covariance differences.

1.12 Cross-layer subspace coupling

Per-module subspaces can be estimated independently. To reduce overfitting and parameter count, modules with compatible activation dimensions can share a global basis

$$V_{\text{gl}} \in \mathbb{R}^{d \times K}.$$

A module-specific subspace can then be represented as

$$V_\ell = V_{\text{gl}} M_\ell. \quad (54)$$

If V_{gl} is column-orthonormal and V_ℓ is required to remain column-orthonormal, then M_ℓ should satisfy

$$M_\ell^\top M_\ell = I. \quad (55)$$

This construction applies directly only to modules sharing the same activation dimension d . For modules with different dimensions, separate global bases or learned mappings into a common latent space are required.

1.13 Practical feasibility

The full covariance implementation requires $\mathcal{O}(d^2)$ storage and can require $\mathcal{O}(Nd^2)$ work for N activation samples. This is practical for small models and selected layers, but becomes expensive for large language models.

The scalable implementation should therefore use covariance–vector products or streaming low-rank PCA, requiring approximately $\mathcal{O}(dK)$ persistent subspace storage.

The low-rank correction itself is considerably cheaper. Once the domain subspace V has been obtained, the matrix

$$A = EV$$

has only K columns, and its compact SVD yields the correction factors without constructing the full projected error.

A useful diagnostic before deploying the correction is the fraction of quantization-error energy captured by the selected subspace:

$$r = \frac{\|EVV^\top\|_F^2}{\|E\|_F^2}. \quad (56)$$

For comparison against a random K -dimensional subspace, one can measure the corresponding ratio

$$r_{\text{random}}.$$

If

$$r \gg r_{\text{random}},$$

the domain-derived subspace is capturing unusually large amounts of quantization error. This provides a stronger sanity check than activation-domain separation alone.

Finally, the correction should be evaluated against the following baselines:

1. the original full-precision model;
2. the NF4-quantized model without correction;
3. a random low-rank correction;
4. a domain-derived low-rank correction;
5. a directly optimized low-rank correction.

The central empirical criterion is whether a small number of bfloat16 parameters can recover a substantial fraction of the target-domain performance lost under NF4 while providing substantially less benefit on the comparison domain.